

Every process function purifies to a unitary process

Draft. Not refereed.

Theorem. The source-and-sink permutation unitary $U|o,e\rangle = |w(o)+e, o\rangle$ is a valid unitary process whenever w is a process function.

Lemma. If s differs from s' , some party disagrees in the output bit and agrees in the function value. Otherwise one local function fixes both strings.

Cancellation. The gram matrix factors across parties.

Off the diagonal the lemma supplies a party whose shift is zero and whose rows differ, so that factor is an inner product of distinct rows and vanishes.

Instruments follow by Stinespring.

Novelty.

Baumeler et al. 2019 left the quantum step open.

Araujo et al. 2017 quantized Lugano only.

Daher Ahmed and Kunjwal, arXiv:2610.00579,
30 September 2026, conjectured the general lift
and proved it under extra sufficient conditions.

This argument does not use those conditions.

The process-function clause is the cancellation.

The forbidden double fixed point is the forbidden
off-diagonal. That identification is the novel step.

Full text: paper/technical.md and paper/essay.md